

SEMESTER PROJECT

Prescriptive Academic Intelligence System

A Hybrid Data Mining Approach to Student Performance Prediction & Targeted Intervention

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BS Data Science | 5th Sem

COURSE INFO

Data Mining

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Project Overview

⚠ Problem Statement

Predicting student success is often reactive. The challenge is to build a hybrid model that can simultaneously:

- ✓ Estimate a student's final exam score (**Regression**).
- ✓ Classify pass/fail status based on thresholds (**Classification**).
- ✓ Identify significant influencing factors (\$x1\$ – \$x7\$) contributing to performance.

🎯 Advanced Project Goal

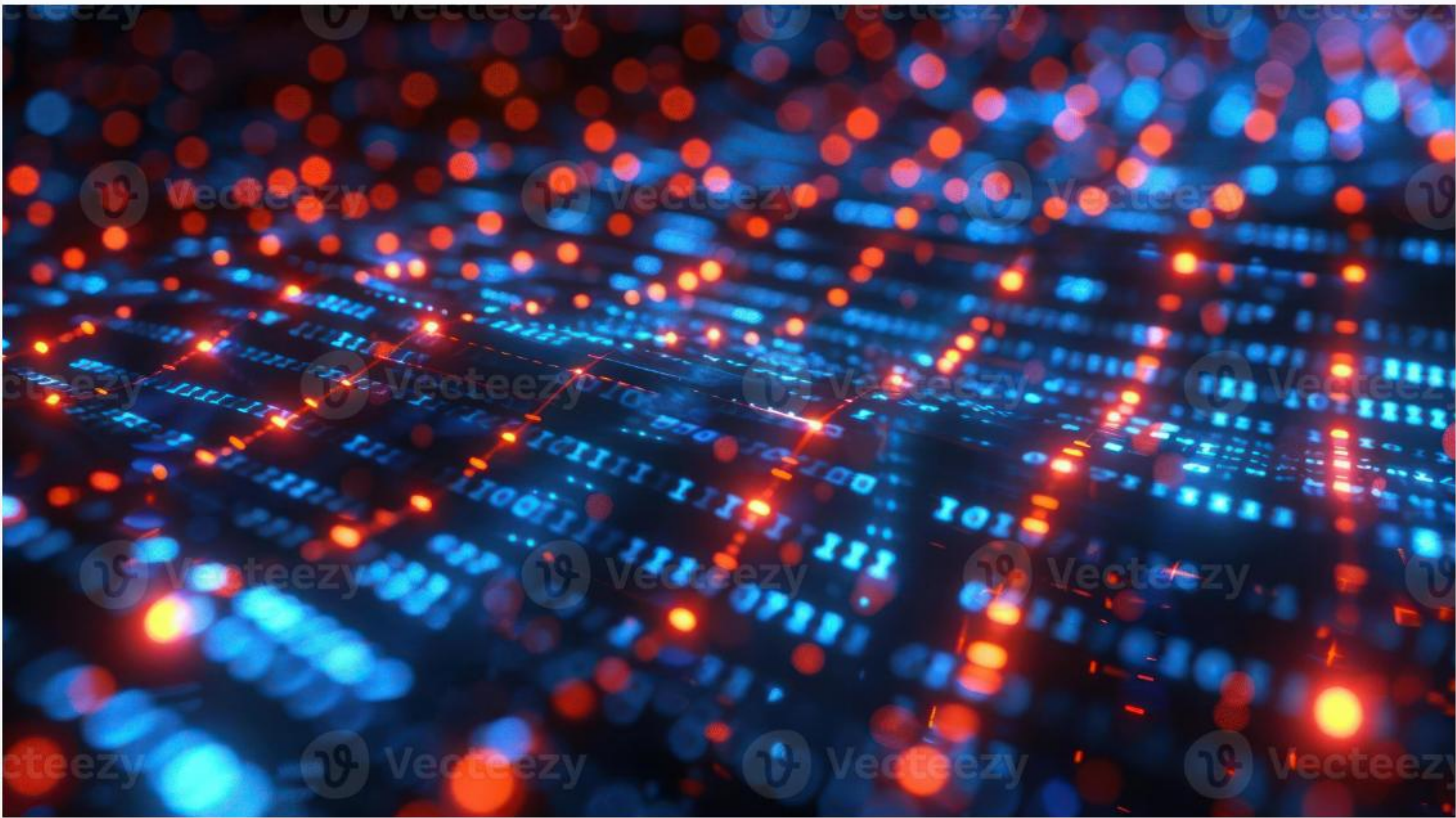
To move beyond simple prediction to **Prescriptive Intelligence**.

We aim to develop a system that delivers actionable, interventional insights. The goal is not just to say "Student A will fail," but to answer: *"How much must score X_i improve to shift the predicted outcome to a higher category?"*

Dataset Overview

Source: Kaggle Students' Academic Performance Dataset

Type	Variable Name	Description
Features (X)	\$x1, x2, ... x7\$	Component scores representing quizzes, assignments, and mid-terms.
Target 1 (Y)	Total Score	Continuous variable (0-100) for Regression tasks.
Target 2 (Y)	Remarks	Categorical variable (1-5) representing performance tiers.



Gap Analysis & Solution



The Gap: Lack of Prescription

Standard data mining projects are **Descriptive**. They identify correlations (e.g., "High study time = High Score") but fail to provide specific, quantified instructions for improvement.

The Solution: Interventional AI

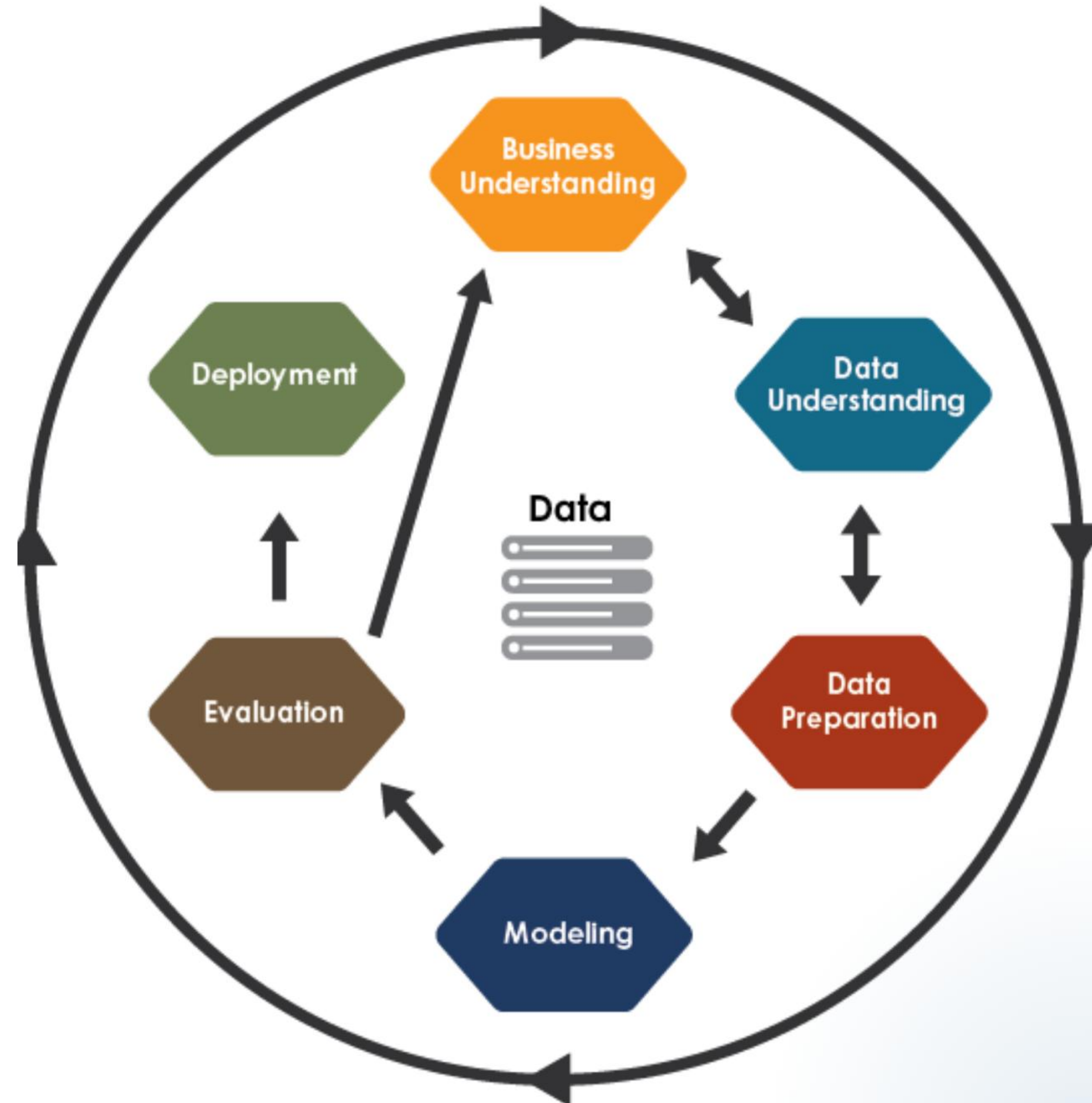
We implement a **Coefficient-Based Counterfactual Simulation**.

- ✓ **Interventional:** Simulates "what-if" scenarios.
- ✓ **Actionable:** Generates rules like "Increase x_7 by 5 points to pass."
- ✓ **Robust:** Uses a high-accuracy hybrid modeling approach.

Methodology

We followed the **CRISP-DM** (Cross-Industry Standard Process for Data Mining) lifecycle to ensure a rigorous and structured approach.

- ✓ **Business Understanding:** Defining the hybrid and prescriptive goals.
- ✓ **Data Prep:** Cleaning, Scaling, and Engineering features.
- ✓ **Modeling:** Training LR, RF, and XGBoost models.
- ✓ **Evaluation:** Comparing metrics (Accuracy, RMSE).
- ✓ **Deployment:** Generating prescriptive rules.



Data Preparation & Feature Engineering



Standard Scaling

Applied `StandardScaler` to all input features (x_1 – x_7). This normalizes the data distribution, which is critical for distance-based models like Logistic Regression.



Consistency Score

Calculated the Standard Deviation of a student's component scores. High variance indicates inconsistent performance across different subjects.



Bottleneck Score

Identified the minimum score across components. The lowest score often acts as the single largest barrier limiting a student's overall grade.

Hybrid Modeling Strategy

☰ Classification Task

Target: Remarks (1–5)

Goal: Maximize Weighted F1-Score & Accuracy.

Algorithms:

- ✓ Logistic Regression (Baseline)
- ✓ Random Forest Classifier
- ✓ XGBoost Classifier

📈 Regression Task

Target: Total Score (0–100)

Goal: Minimize RMSE (Root Mean Square Error).

Algorithms:

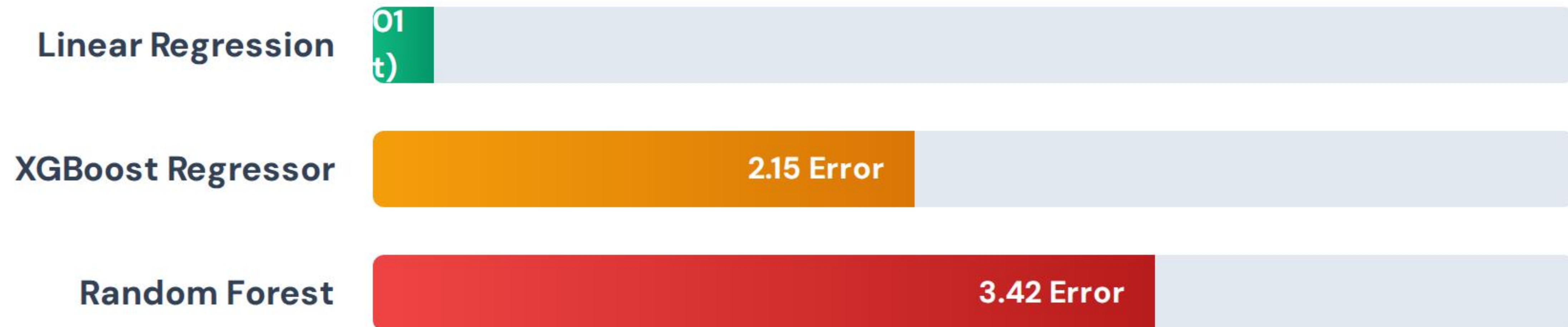
- ✓ Linear Regression
- ✓ Random Forest Regressor
- ✓ XGBoost Regressor

Results: Classification Performance



***Key Finding:** Logistic Regression outperformed complex tree-based models, achieving near-perfect accuracy. This indicates a strong linear relationship between component scores and the final grade.*

Results: Regression Performance (RMSE)



For regression, Linear Regression achieved an almost zero error rate. This confirms that the Total Score is a linear summation of features x_1 – x_7 .

Prescriptive Analysis: Key Drivers

x7

Highest Leverage Feature

Global Insight

By analyzing the scaled coefficients of our Logistic Regression model, we identified **Component \$x7\$** as the most critical driver for high performance.

Improving this specific score yields the highest return on investment for a student's final grade.

Local Insight: Counterfactual Simulation

Before Intervention

Student ID: #3

Predicted Status: Remarks 2 (Failing)

Identified Weakness: Low $\times 7$ score.

After Intervention

Action: Increase $\times 7$ by 5 points.

New Prediction: **Remarks 3 (Passing)**

Result: Successful Category Shift.

This simulation proves the model's ability to provide personalized, actionable advice rather than just a prediction.

Conclusion & Future Work

- ✓ **Objective Met:** Built a near-perfect hybrid predictor using Logistic Regression.
- ✓ **Gap Closed:** Delivered a "Prescriptive Intelligence" system that generates if-then rules for educators.
- ↗ **Future Work:** Deploy as a real-time API and implement Multi-Task Learning (MTL) Neural Networks for further optimization.

Questions?

Thank you for your attention.

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Image Sources



https://static.vecteezy.com/system/resources/previews/055/370/625/large_2x/abstract-digital-background-with-glowing-red-and-blue-data-points-creating-a-vibrant-matrix-effect-photo.jpg

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